

Statistical Learning with High-dimensional Data



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The Gaussian Mixture Model and the EM algorithm:

This approach has clearly a more sound but also deeper theoretical assumptions, but it provides also more flexibility and more features.

Definition: A mixture model is a statistical

model where the p.d.f. can be written as follows:

$$p(x) = \sum_{k=1}^K \pi_k p_k(x)$$

where: $\pi_k \in [0,1]$

- $\sum_k \pi_k = 1$

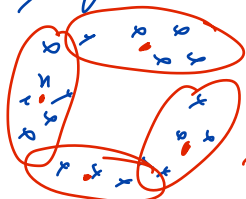
- $p_k(x)$ is a p.d.f.

Among all possible mixture models we can generate,
we can for instance consider:

continuous $\left\{ \begin{array}{l} \bullet p(x) = \underline{0.1} \underline{N(x; 0, 1)} + \underline{0.7} \cdot \underline{N(x; -1, 1)} + \underline{0.2} \underline{U_{[0,1]}(x)} \\ \bullet p(x) = \underline{0.7} \underline{N(x; 0, 1)} + \underline{0.3} \underline{St(x; 5)} \end{array} \right.$

count $\left\{ \bullet p(\underline{x}) = \underline{0.5 P(x; 1)} + \underline{0.5 P(x; 5)} \right.$

Rule: mixture models can be very flexible because you
can also vary the nb
of components



there, a mixture
of 4 Gaussians is
fitting well this
non-Gaussian dataset.

In practice, the mixture model which is the most popular is the Gaussian mixture model (GMM)

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k) *$$

where μ_k and Σ_k are the means and covariances of the k^{th} component.

In order to perform clustering with this model, we need to introduce a latent (non-observed) variable Y indicating the cluster membership, and such that $P(Y=k) = \pi_k$.

Then, the GMM can be rewritten:

$$\begin{cases} \underline{Y} \sim \underline{\mathcal{M}(\pi_k)} \\ \underline{X|Y=k} \sim \underline{\mathcal{N}(\mu_k, \Sigma_k)} \end{cases} \quad *$$

$\Rightarrow$ clustering with the GMM reduces to estimate from data the mixture parameters $\Theta = \{\pi_1, \dots, \pi_K, \mu_1, \dots, \mu_K, \Sigma_1, \dots, \Sigma_K\}$ and the latent variable Y .

Unfortunately, maximizing the likelihood associated to this model is not tractable and we have to use an iterative algorithm: the EM algorithm.

The Expectation-Maximization (EM) algorithm:

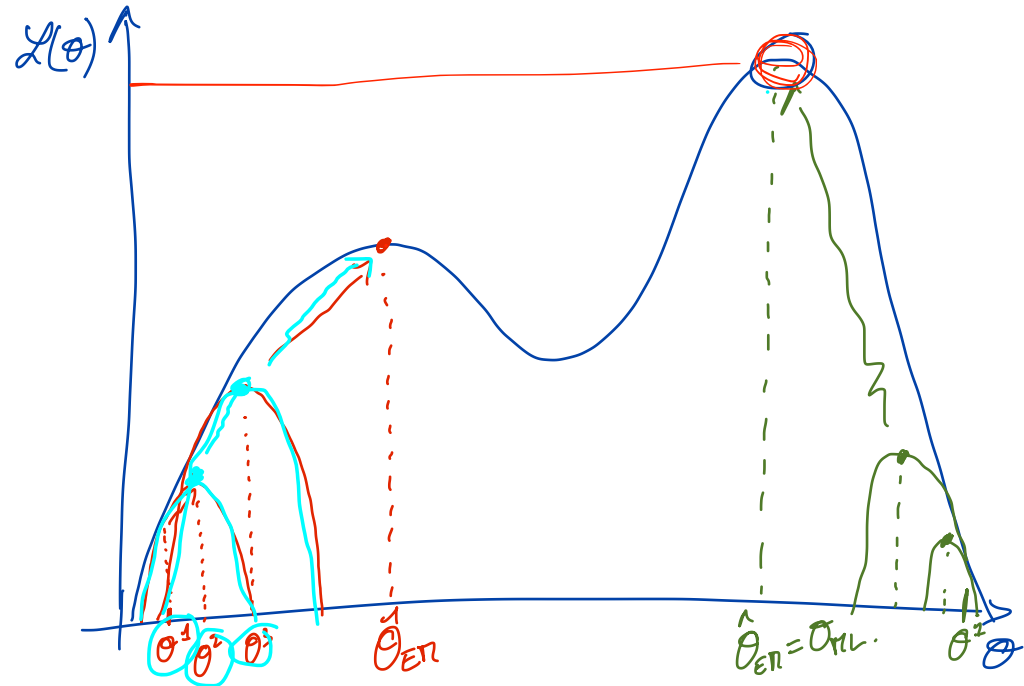
Dempster, Laird and Rubin (1977) proposed the EM algorithm to optimize the likelihood of models with latent variables.

The idea of the EM algorithm is to first suppose that Y is known, and get values for θ ; then given this value of θ , deduce values for Y .

So, the EM algorithm alternates:

- the E step: we calculate $E[Y|\theta^*, X]$

- the M step: maximize the conditional expected complete likelihood to get θ : $\theta^* = \underset{\theta}{\operatorname{argmax}} E[\log p(X, Y|\theta; \theta^*, X)]$



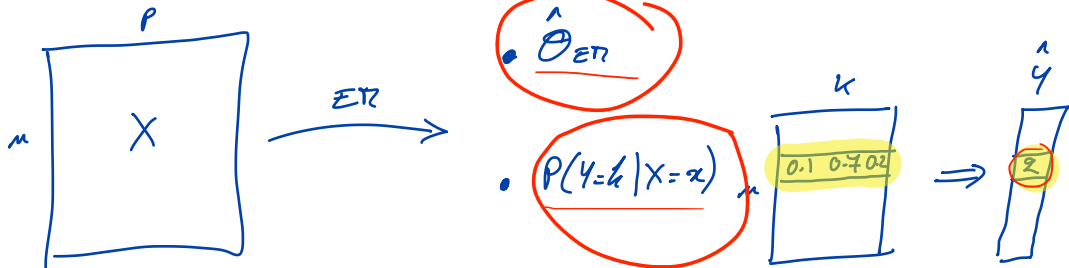
Theorem (DLR77 + Wu79): The sequence of $(\theta_j^{(q)})_j$ generated by the EM algorithm is converging toward a local maximum of the log-likelihood.

A way to avoid being trapped in a local max of the likelihood is to do several initializations of the EM instead of a unique one:

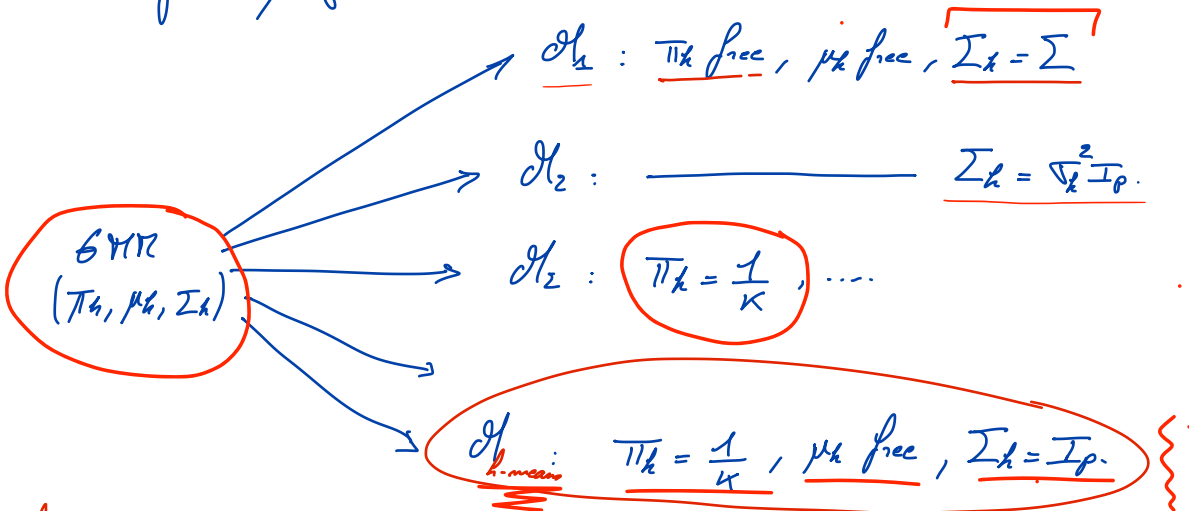
$$\left. \begin{array}{l} \theta_1^{(1)} \xrightarrow{\text{EM}} \hat{\theta}_1 + L(\hat{\theta}_1) \\ \vdots \\ \theta_B^{(1)} \xrightarrow{\text{EM}} \hat{\theta}_B + L(\hat{\theta}_B) \end{array} \right\} \hat{\theta}_{\text{EM}} \text{ associated with the highest likelihood } L(\cdot)$$

Once the ETR algorithm has been learned for the data at hand, it is possible to output:

- $\hat{\Theta}_{ETR}$ = $\{ \underbrace{\hat{\pi}_1, \dots, \hat{\pi}_k}_{\text{probabilities of the groups}}, \underbrace{\hat{\mu}_1, \dots, \hat{\mu}_k}_{\text{cluster means}}, \underbrace{\hat{\Sigma}_1, \dots, \hat{\Sigma}_k}_{\text{covariance matrices of clusters}} \}$
- $y_i = \arg \max_{k=1 \dots k} \underbrace{P(Y=k | X=x_i)}_{\text{Bayes}} \propto \underbrace{\frac{1}{\pi_k} N(x_i, \hat{\mu}_k, \hat{\Sigma}_k)}_{\text{Gaussian distribution}}$

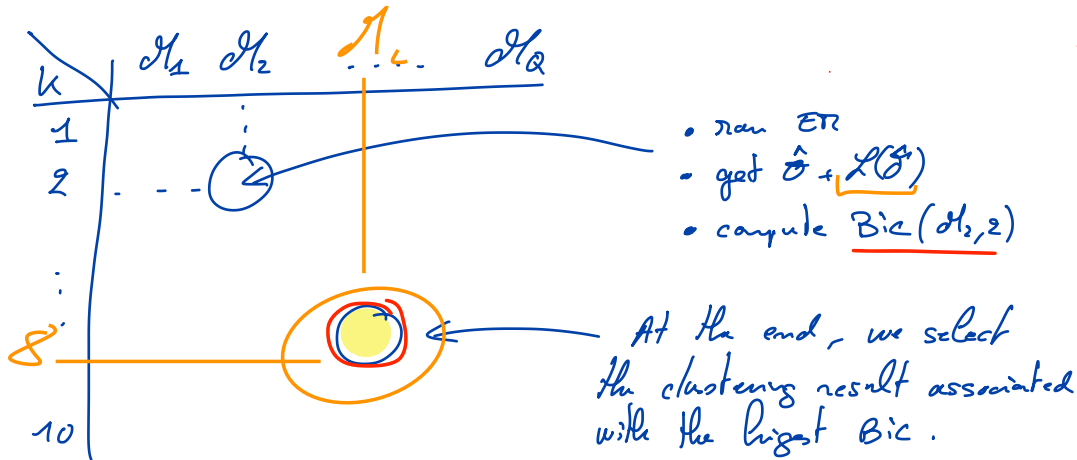


Another interest of the GTR is its ability to be extended as a family of sub-models:



h -means is a basic version of the ER algorithm for a very constrained GTR.

A final advantage of this approach is to be able to choose the most appropriate nb of groups, k , using model selection criteria:



Remember that the equation for $\text{Bic}(\mathcal{M})$ is as follows:

$$\text{Bic}(\mathcal{M}) = \log \mathcal{L}(\hat{\theta}_{\mathcal{M}}) - \underbrace{\frac{V(\mathcal{M})}{2}}_{\log n} \underbrace{\log n}_{\text{red squiggle}}$$

where $V(\mathcal{M})$ is the nb of free scalar parameters in $\mathcal{M}$.

Exercise: write down the value of $V(\text{GMK})$

$$\underbrace{V(\text{GMK})}_{\text{orange bracket}} = \left[\underbrace{(K-1)}_{\downarrow} + \underbrace{K \cdot p}_{\downarrow} + \underbrace{K p(p-1)/2}_{\text{orange oval}} \right]$$
$$\# \{ \underbrace{\pi_1, \dots, \pi_K}_{\text{blue underline}}, \underbrace{\mu_1, \dots, \mu_K}_{\text{red oval}}, \underbrace{\Sigma_1, \dots, \Sigma_K}_{\text{red oval}} \}$$

Summary about GMM/EM:

- ⊕ a statistical model, with sound theoretical foundations
- ⊕ the value of K and the submodel can be found using model selection.
- ⊕ a knowledge of the clustering risk (thanks to the posteriors)
- ⊕ mixture parameters are possible to interpret.
- ⊖ the theory is quite complex
- ⊖ } the GMM with full covariance matrices is not working well in HD spaces.

Outline

1. Introduction
2. Reminder of descriptive statistics
3. The learning process
4. Supervised learning
5. Unsupervised learning
6. Learning in high-dimensions
- ~~7. Dimension reduction~~
8. Clustering in high dimensions

Learning in high-dimensions

Learning in high-dimensions is one of the most important problems nowadays:

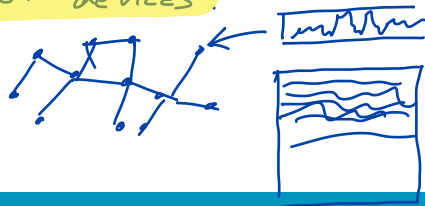
Due to the evolution of sensors in the last 20-30 years, we are now able to record a lot of data with high precision = high. dimensional.

→ image analysis.

→ medical imaging.

→ signal processing

→ IOT devices.



The curse of dimensionality

The **curse of dimensionality**:

- this term was first used by R. Bellman in the introduction of his book “Dynamic programming” in 1957:

All [problems due to high dimension] may be subsumed under the heading “the curse of dimensionality”. Since this is a curse, [...], there is no need to feel discouraged about the possibility of obtaining significant results despite it.

- he used this term to talk about the difficulties to find an optimum in a high-dimensional space using an exhaustive search,
- in order to promote dynamic approaches in programming.

The curse of dimensionality

In the **mixture model** context:

- the building of the data partition mainly depends on:

$$H_k(x) = -2 \log(\pi_k f(x, \theta_k)),$$

- model **Full-GMM**:

$$H_k(x) = (x - \mu_k)^t \Sigma_k^{-1} (x - \mu_k) + \log(\det \Sigma_k) - 2 \log(\pi_k) + \gamma.$$

Consequently:

- it is necessary to invert Σ_k which have a number of parameters proportional to p^2 ,
- if n is small compared to p^2 , the estimates of Σ_k are ill-conditioned or singular and it will be difficult or impossible to invert Σ_k .

$$n < p \rightarrow \det(\Sigma_k) = 0$$

$$\begin{aligned} P(y=h|X, \theta) \\ \propto P(y=h) \prod_{i=1}^n N(x_i | y=h, \theta) \end{aligned}$$

The curse of dimensionality

From the estimation point of view:

- let us consider the **normalized trace** $\tau(\Sigma) = \text{tr}(\Sigma^{-1})/p$ of the inverse covariance matrix Σ^{-1} of a **multivariate Gaussian distribution** $\mathcal{N}(\mathbf{0}, \Sigma)$,
- the **estimation of τ** from a sample of n observations $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ $\sim \mathcal{N}(\mathbf{0}, \Sigma)$ conduces to:

$$\hat{\tau}(\hat{\Sigma}) = \tau(\hat{\Sigma}) = \frac{1}{p} \text{tr}(\hat{\Sigma}^{-1}), \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n} \bar{\mathbf{X}}^t \bar{\mathbf{X}}$$

$$E[\tau(\hat{\Sigma})] = \left(1 - \frac{p}{n-1}\right)^{-1} \tau(\Sigma).$$

- consequently**, if the ratio $p/n \rightarrow 0$ when $n \rightarrow +\infty$, then $E[\tau(\hat{\Sigma})] \rightarrow \tau(\Sigma)$, (unbiased estimator)
- however**, if the dimension p is comparable with n , then $E[\tau(\hat{\Sigma})] \rightarrow c\tau(\Sigma)$ when $n \rightarrow +\infty$, where $c = \lim_{n \rightarrow +\infty} p/n$.

$$E[\tau(\hat{\Sigma})] = (1 - c)^{-1} \tau(\Sigma) \Rightarrow \text{(biased estimator)}$$

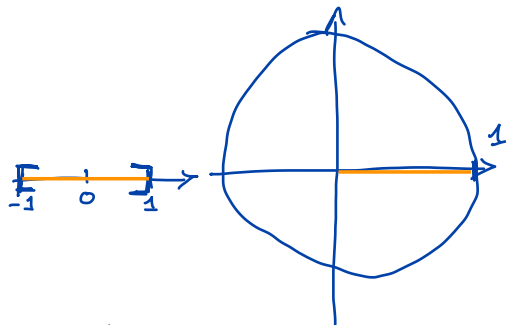
The blessings of dimensionality

in R: gamma()

As Bellman thought:

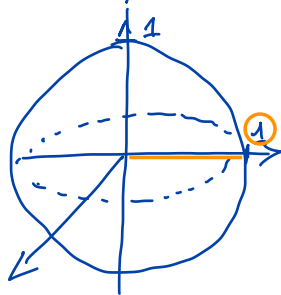
- all is not bad in high-dimensional spaces (hopefully!)
- there are interesting things which happen in high-dimensional spaces.

First example: volume of the unit sphere is $V(p) = \frac{\pi^{p/2}}{\Gamma(p/2+1)}$,
Unit hyper-sphere

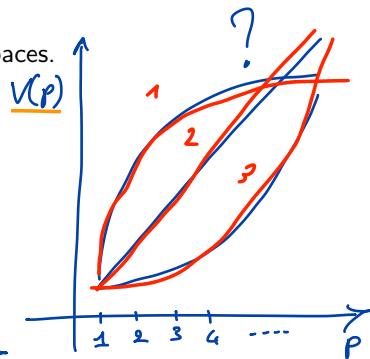


p=1

p=2



p=3



p=4 ... ∞

The blessings of dimensionality

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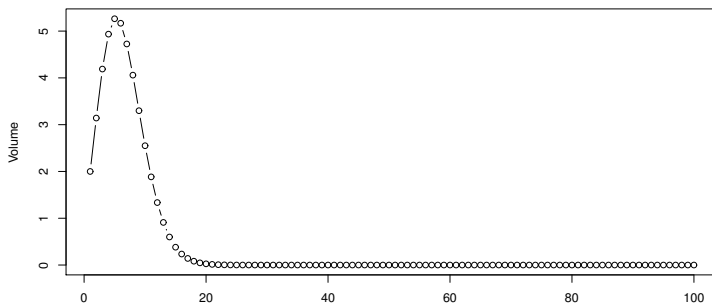


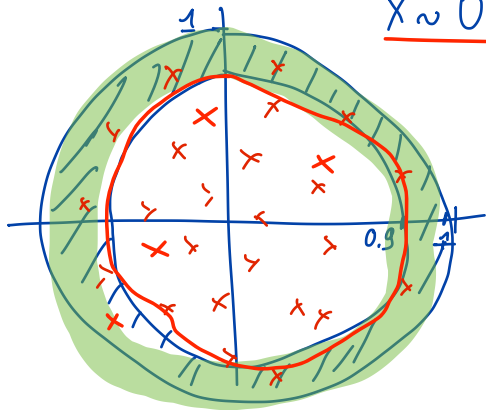
Fig. Volume of a sphere of radius 1 regarding to the dimension p .

The blessings of dimensionality

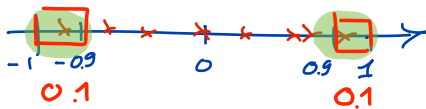
Second example: probability that a uniform variable on the unit sphere belongs to the shell between the spheres of radius 0.9 and 1 is

$$P(X \in S_{0.9}(p)) = 1 - 0.9^p \xrightarrow{p \rightarrow \infty} 1$$

$$X \sim U(S_1)$$



$$P(X \in S_{0.9}) = 0.1$$

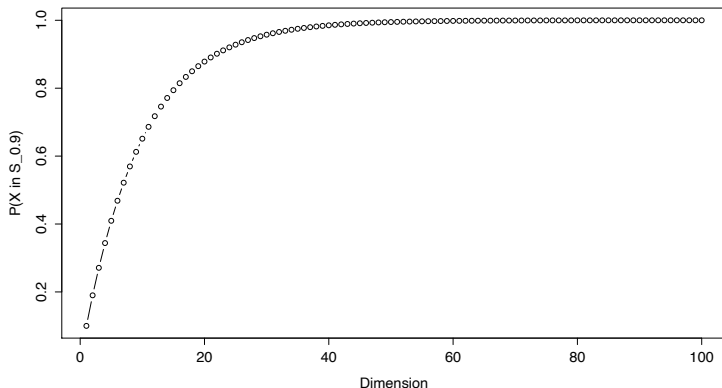


The blessings of dimensionality

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$$P(X \in S_{0.9}(p)) = 1 - 0.9^p \xrightarrow{p \rightarrow \infty} 1$$

$\Rightarrow$ this is a mark that HD spaces are almost empty.



$\hookrightarrow$ the data live in low-dimensional subspaces.

Fig. Probability that X belongs to the shell $S_{0.9}$ regarding to the dimension p .

The blessings of dimensionality

Third example:

- since high-dimensional spaces are almost empty,
- it should be easier to separate groups in high-dimensional space with an adapted classifier,
- a way to observe this is to look at the Bayes classifier behaviour.

The blessings of dimensionality

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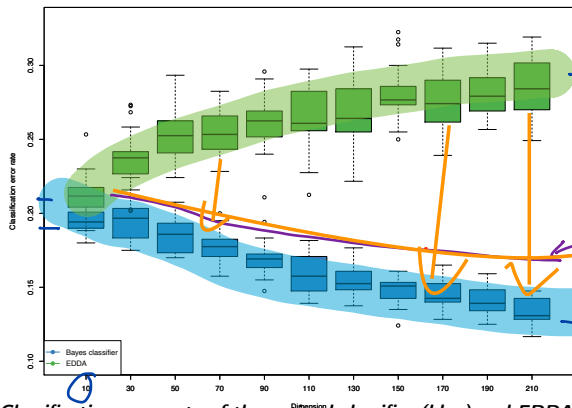


Fig. Classification error rate of the optimal classifier (blue) and EDDA (green) versus the data dimension on simulated data.

Classical ways to avoid the curse of dimensionality

Dimension reduction:

- the problem comes from that p is too large,
- therefore, reduce the data dimension to $d \ll p$,
- such that the curse of dimensionality vanishes!

Regularization:

- the problem comes from that parameter estimates are unstable,
- therefore, regularize these estimates,
- such that the parameter are correctly estimated!

Parsimonious models:

- the problem comes from that the number of parameters to estimate is too large,
- therefore, make restrictive assumptions on the model,
- such that the number of parameters to estimate becomes more “decent”!

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Dimension reduction

A common phantasm about dimension reduction:

- believe that dimension reduction helps for classification,
- **this is not true** because, most of the time, dimension reduction implies an information loss which would be discriminative.

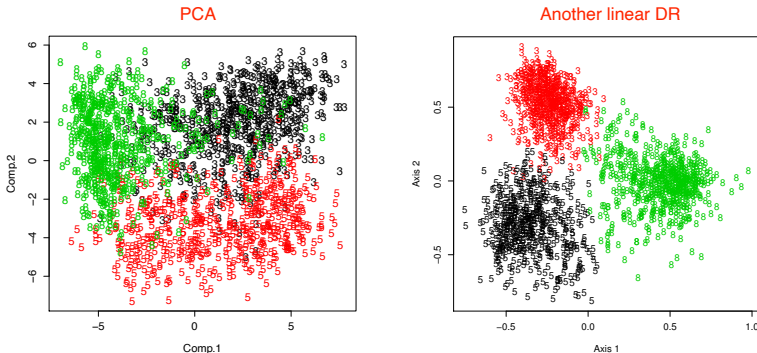


Figure: Projection of the 256-dimensional USPS data with PCA (left, unsupervised) and FDA (right, supervised).

Regularization: (the ridge regularization)

The pb is to invert $\Sigma_1 \rightarrow (\hat{\Sigma}_k)^{-1}$

A simple way to avoid this pb and regularize is to add a small positive value on the diagonal of $\hat{\Sigma}_k$:

$$\left(\hat{\Sigma}_k^{-1} \right) = \left(\hat{\Sigma}_k + \lambda \mathbf{I}_p \right)^{-1}$$

where $\lambda > 0$
(usually $\lambda \sim \underline{1e^{-3}}$)

Using parsimonious models:

the GMM is over-parameterized (too many parameters to estimate)

↳ make some assumptions to reduce this nb of parameters to estimate.

Parsimonious models for GMM

p -means
 $V(\mathcal{O}) = k p$
 $= 400$

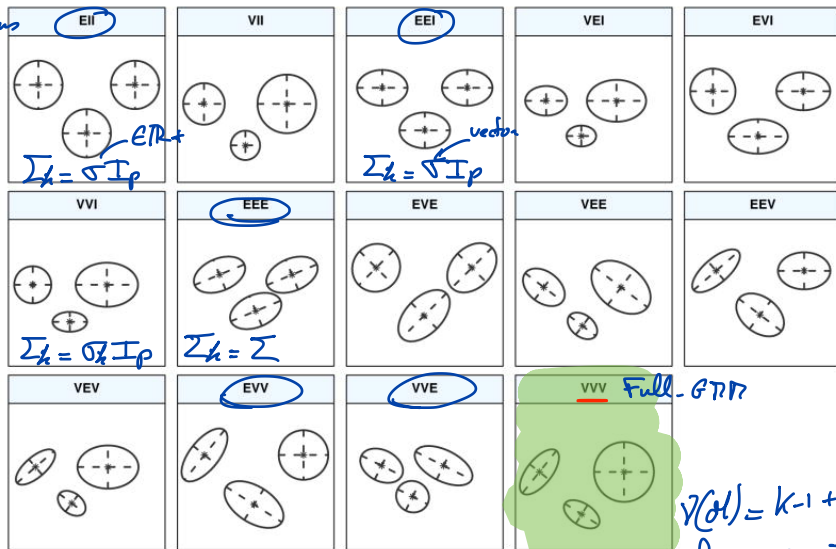


Figure: The parsimonious models of Mclust.

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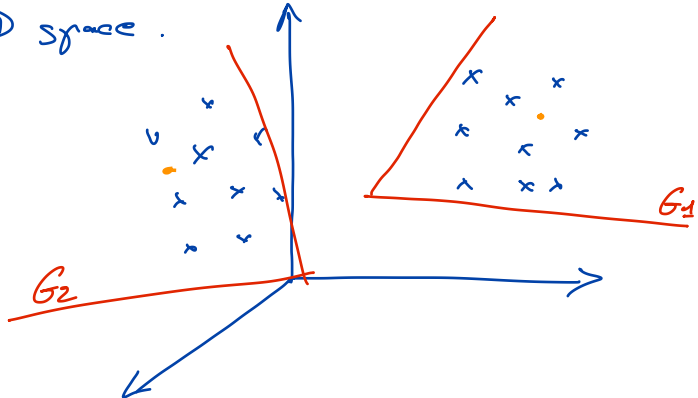
The solutions

In fact, the good idea to face the curse of dimensionality is to do dimension reduction, regularization and submodels at the same time.

⇒ classification and clustering in low-dimensional subspaces, (with variable selection if needed).

The general idea of subspace clustering:

Those methods exploit the "empty space phenomenon" by assuming that the data are living in low-dimensional and cluster-specific subspaces of the original HD space.



The mixture of PPCA

The mixture of PPCA was proposed by Tipping & Bishop (1999) and assumes that the mixture incorporates an internal and group-specific dimension reduction through PPCA.

The model of **M**PPCA:

$$\bullet \quad Y \sim \mathcal{M}(1; \pi) \Rightarrow P(Y=k) = \pi_k$$

$$\bullet \quad Z | Y=k \sim N(z; \mu_k, Id) \quad \text{where } \mu_k \in \mathbb{R}^d \text{ with } d < p$$

$$\bullet \quad E | Y=k \sim N(e; 0, \Sigma_k^{-1} I_p)$$

$$\bullet \quad X = U_k^T Z + E \quad \text{if } Y=k$$

we recognize the
PPCA model
conditioned on Y .

The mixture of PPCA

$$X = U_k^t Z + \epsilon$$

← additive noise in $\mathbb{R}^p$

the latent representation of the data in $\mathbb{R}^d$

the projection matrix of size $p \times d$

It is possible to write the marginal distribution of X :

$$p(X) = \sum_{h=1}^K \pi_h \mathcal{N}(x; \mu_h, U_h^t U_h + \sigma_h^2 I_p)$$

a specific covariance structure

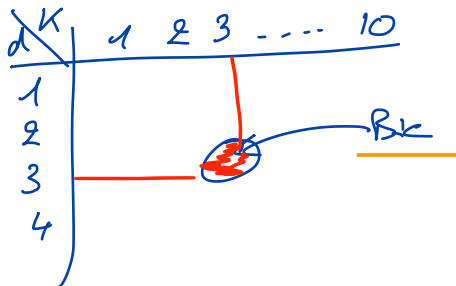
$\Rightarrow$ we recognize a GMM with a specific covariance : $\Sigma_h = U_h^t U_h + \sigma_h^2 I_p$

The mixture of PPCA

The estimation of the MPPCA model can be done with the EM algorithm, for a given value of K and d .

The choice of K and d can be done using model selection (using BIC)

$$(\hat{K}^*, \hat{d}^*) = \underset{K, d}{\operatorname{argmax}} \quad \underline{\operatorname{Bic}(K, d)}$$



Calculation of BIC for MPPCA:

$$\text{Bic}(\text{MPPCA}) = \underline{\log(\mathcal{L}(x; \hat{\theta}))} - \frac{\boxed{V(d)}}{2} \log(n)$$

$$\text{where } V(d) = \underbrace{K-1}_{\text{proportions}} + \underbrace{Kd}_{\text{means}} + K \underbrace{\left[\frac{p(d+1)}{2} + 1 \right]}_{\text{covariance}}$$

Rank: We clearly see here that MPPCA is parsimonious compared to the full GMD.

Rank: it is worth remarking that MPPCA generalizes the GMD since it can recover it when $d = p-1$.

HDDC (Bouveyron, Girard, Schmid, 2006)

HDDC (High-dimensional data clustering) generalizes the idea of MPRCA by relaxing the constraint on d and by allowing other constraints on model parameters:

- $Y \sim \mathcal{H}(1; \pi)$
- $Z | Y=k \sim N(z; \mu_k \mathbb{I}_{d_k})$ where $\mu_k \in \mathbb{R}$ and d_k
- $E | Y=k \sim N(e; 0, \sigma_k^2 \mathbb{I}_p)$
- $X = U_k^T Z + E$ if $Y=k$.

The most general model of HDDC:

$$[\underline{a_{kj}} \quad \underline{b_k} \quad \underline{Q_k} \quad \underline{d_k}] = \underline{\text{MPPCA}} + \text{free } d_k$$

$\uparrow$ $\|u_{kj}\|$ $\uparrow$ $\underline{\sigma_k^2}$ $\uparrow$ U_k

$\rightarrow [\underline{a_{kj}} \quad \underline{b} \quad Q_k \quad \underline{d_k}] \rightarrow$ variance of the noise common between groups

28 models
in the
family.

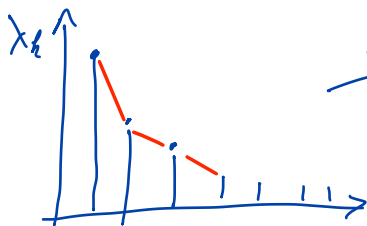
$[\underline{a_k} \quad b_k \quad Q_k \quad d_k] \rightarrow$ variance of the data is spherical

$[\underline{a_{kj}} \quad \underline{b_k} \quad Q_k \quad \underline{d}] \rightarrow \underline{\text{PPCA}}$

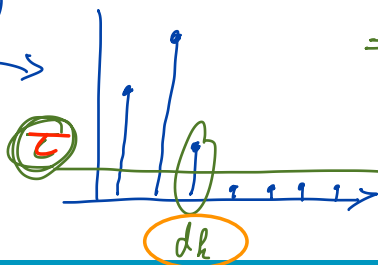
$[\underline{a} \quad \underline{b} \quad Q \quad \underline{d}] \rightarrow \underline{\text{Kmeans} + \text{PCA}}$

For choosing the dimension d_k with BIC, we need a way to avoid the combinatorial exploration of all combinations:

for each group, we apply the score-test of Cattell on the eigenvalue score of $\hat{\Sigma}_k$



diff $\rightarrow$



$\Rightarrow$ we reduced the choice of all d_k to the choice of a single parameter τ by BIC

Bic will be used to select:

- one of the submodels of the family

- $k \in [1, \dots, k_{\max}]$

- $\tau \in [0, 1]$ if τ is close to 1 $\Rightarrow$ $d_h = 1 \quad \forall h$
 $\tau \xrightarrow{\hspace{1cm}} 0 \Rightarrow$ $d_h \rightarrow \rho \quad \forall h$.

-
- HDDC is implemented in R in the HDDclassif package
 - HDDclassif also implements the supervised version HDDA

Summary on MPPCA / HDPC :

- ⊕ a GMR adapted to HD data
- ⊕ no problem with the curse of dimensionality
- ⊕ all advantages of GMR.
- ⊖ it is not easy to visualize because of the different subspaces.